

Data Science

Assignment – 8

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1. Data-driven organizations

Q. Describe, using your own words, the key points that a data-driven should possess according to Thusoo and Sarma. Write approx. 1200 words.

Solution:

According to Thusoo and Sarma, a data-driven organization should possess three key points:

1. A culture in which everyone buys into the idea of using data to make business decisions
2. An organizational structure that supports a data-driven culture
3. Technology that supports a data-driven culture and makes data self-service

1. A culture in which everyone buys into the idea of using data to make business decisions:

The right technology and processes are necessary, but creating a data-driven culture also calls for a change in thinking and behavior. Everyone in a company that is data-driven, from top executives to entry-level workers, is aware of the importance of using data to guide decision making. This can entail analyzing data to put theories to the test, evaluate the success of plans, and spot areas for improvement. Transparency, a readiness to question presumptions, and an openness to embracing new insights—even if they run counter to conventional wisdom—are additional requirements for a data-driven culture.

To foster a data-driven culture, an organization should:

1. Encourage employees to ask questions and seek out data to inform their work.

2. Provide training and resources to help employees understand and use data effectively.
3. Reward employees for using data to drive decision making and innovation.
4. Encourage a culture of continuous learning and experimentation, where employees are encouraged to test hypotheses and learn from data.

A dedication to data-driven decision making at all organizational levels and support from the leadership are necessary for the development of a data-driven culture. It also necessitates a readiness to respond to new data-driven insights and welcome change.

There can be few potential roadblocks while implementing a data culture in an organization:

The most frequent barrier to establishing a self-service data culture is resistance from the group of individuals. They have served as the link between the data and the users. According to them, we can't provide everyone access to the data. Because both security and compliance issues might exist. But, with the help of technology we can resolve them. Another potential issue is that since you make your data repositories accessible to everyone, you might discover that your infrastructure is insufficient to handle such widespread access. Utilizing infrastructure that can scale efficiently will help you solve this problem.

A clear governance framework that outlines how data will be managed, accessed, and used inside the company is also crucial, as is the establishment of clear roles and responsibilities for tasks linked to data. This could make sure that data is used consistently and effectively throughout the company and that it is safe from misuse or illegal access.

In general, a culture where everyone accepts the concept of utilizing data to make business decisions is one that values openness, transparency, and data-driven decision-making as well as continual learning and development. An organization can build a solid basis for using data to guide decision-making and propel commercial success by building this type of culture.

2. An organizational structure that supports a data-driven culture:

An organization needs a supportive organizational structure for a data-driven culture to thrive. This could entail designing specific data teams, defining roles and responsibilities for tasks involving data, and putting in place data governance procedures to guarantee the security and integrity of data. The organizational structure must also be versatile and flexible in order to respond to shifting priorities and demands.

Of course, you want the best employees working for you. Each data team that is put together will be made up of a variety of people with a variety of skills because each organization, of course, has different demands and objectives. Despite this, there are key things that all data teams, regardless of the industry they serve, share. Clearly defining each team member's tasks and responsibilities will help to prevent friction and inefficiencies.

Identification of significant stakeholders and decision-makers inside the business, as well as the establishment of transparent channels for communication and accountability, are necessary to build an organizational structure that supports a data-driven culture. In order to create data strategy and prioritize data activities, this may entail establishing a data steering committee or other organization. Roles and duties for data-related tasks, such as data management, analysis, and visualization, should be clearly defined. This can aid in ensuring that data is utilized consistently and successfully throughout the company.

The management, access, and usage of data within the firm must also be clearly outlined in a data governance framework. As part of this, processes for managing and updating data may be established along with policies and procedures for data security, privacy, and compliance. An organization can make sure that data is utilized ethically and responsibly and that it is safeguarded from misuse or unauthorized access by setting certain rules.

Overall, an organizational structure that promotes a data-driven culture is one that has clear channels of communication and accountability for data-related tasks and is transparent, adaptable, and accountable. An organization may build a solid basis for leveraging data to guide decision-making and propel commercial success by putting in place this kind of framework.

3. Technology that supports a data-driven culture and makes data self-service:

An organization needs to have the appropriate technology in place for data to be accessible and usable. This could comprise infrastructure to support data processing and storage, as well as tools for data management, analysis, and visualization. Additionally, it is crucial that this technology be simple to use and user-friendly so that staff members may access and utilize data without the need for advanced technical knowledge. Data should be self-service in a data-driven business, allowing employees to access and use information without passing through a central IT department or other gatekeepers. Employees may now make data-driven decisions more quickly and effectively thanks to this.

It's crucial to pick tools and platforms that are user-friendly, scalable, and secure in order to build a technological environment that supports a data-driven culture and enables data self-service. Picking a visualization tool that enables users to make high-quality plots and charts without the need for specialist technical knowledge or choosing a data management platform that is simple to use and connects with other

tools and systems are two examples of how to accomplish this. Establishing precise standards and procedures for handling data, including how it is stored, secured, and accessible, is also crucial. This can support ensuring that information is used morally and sensibly.

Furthermore, it is crucial to take into account the infrastructure required to support data processing and storage. This may entail spending money on sophisticated servers and storage systems, as well as other equipment and software, to support data processing, or selecting for cloud-based solutions to deal with on-site infrastructure. In order for the technology to develop and grow as the organization's data needs change, it is also critical to take its scalability into account.

In general, technology that promotes a data-driven culture and enables data self-service is user-friendly, scalable, and secure and is created to enable staff members to access and use data quickly and effectively. An organization may lay a solid basis for leveraging data to guide decision-making and propel economic success by developing this type of technological environment.

Therefore, a data-driven organization should possess a culture in which everyone buys into the idea of using data to make business decisions, an organizational structure that supports a data-driven culture, and technology that supports a data-driven culture and makes data self-service. By establishing these three points, an organization can create a strong foundation to use data for informing decision-making and drive business success.